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United States Patent	12416821
Kind Code	B2
Date of Patent	September 16, 2025
Inventor(s)	Bian; Yusheng

Plasmonic photonic structures including a layer that exhibits an electric-field-induced Pockels effect

Abstract

Plasmonic photonic structures that include a layer that exhibits an electric-field-induced Pockels effect and methods of forming such structures. The structure comprises a waveguide core on a substrate, a first layer that has an overlapping relationship with the first waveguide core, and a second layer that has an overlapping relationship with the first waveguide core and the first layer. The first layer comprises a metal, and the second layer comprising a material that exhibits an electric-field-induced Pockels effect.

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Appl. No.:	18/212437
Filed:	June 21, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240427177 A1	Dec. 26, 2024

Publication Classification

Int. Cl.: G02B6/122 (20060101); G02F1/015 (20060101); G02F1/035 (20060101); G02F1/313 (20060101)

U.S. Cl.:

CPC **G02F1/0154** (20210101); **G02B6/1226** (20130101); **G02F1/035** (20130101);
G02F1/3132 (20130101); G02F2201/063 (20130101); G02F2202/20 (20130101);
G02F2203/10 (20130101)

Field of Classification Search

CPC: G02F (1/0154); G02F (1/035); G02F (1/3132); G02F (2201/063); G02F (2202/20); G02F (2203/10); G02B (6/1226)

USPC: 385/1-3; 385/9; 385/14; 385/28; 385/30

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Background/Summary

BACKGROUND

(1) The disclosure relates to photonics chips and, more specifically, to plasmonic photonic structures that include a layer that exhibits an electric-field-induced Pockels effect and methods of forming such structures.

(2) Photonics chips are used in many applications and systems including, but not limited to, data

communication systems and data computation systems. A photonics chip includes a photonic integrated circuit comprised of photonic components, such as modulators, polarizers, and optical couplers, that are used to manipulate light received from a light source, such as an optical fiber or a laser.

(3) Thin film lithium niobate (TFLN) may be integrated into a photonic component. Lithium niobate is an electro-optical material characterized by optical properties, such as large second-order nonlinearity, a wide transparency window, and low nonlinear absorption, that are favorable for certain types of photonic components. Photonic components including a TFLN layer may suffer from several disadvantages, such as a large footprint and a limited power handling capability.

(4) Improved plasmonic photonic structures that include a layer that exhibits an electric-field-induced Pockels effect and methods of forming such structures are needed.

SUMMARY

(5) In an embodiment of the invention, a structure for a photonics chip is provided. The structure comprises a waveguide core on a substrate, a first layer that has an overlapping relationship with the first waveguide core, and a second layer that has an overlapping relationship with the first waveguide core and the first layer. The first layer comprises a metal, and the second layer comprising a material that exhibits an electric-field-induced Pockels effect.

(6) In an embodiment of the invention, a method of forming a photonics chip is provided. The method comprises forming a waveguide core on a substrate, forming a first layer that has an overlapping relationship with the waveguide core, and forming a second layer that has an overlapping relationship with the waveguide core and the first layer. The first layer comprises a metal, and the second layer comprises a material that exhibits an electric-field-induced Pockels effect.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The accompanying drawings, which are incorporated in and constitute a part of this specification, illustrate various embodiments of the invention and, together with a general description of the invention given above and the detailed description of the embodiments given below, serve to explain the embodiments of the invention. In the drawings, like reference numerals refer to like features in the various views.

(2) FIG. 1 is a top view of a structure at an initial fabrication stage of a processing method in accordance with embodiments of the invention.

(3) FIG. 2 is a cross-sectional view taken generally along line 2-2 in FIG. 1.

(4) FIG. 3 is a top view of the structure at a fabrication stage of the processing method subsequent to FIGS. 1, 2.

(5) FIG. 4 is a cross-sectional view taken generally along line 4-4 in FIG. 3.

(6) FIG. 5 is a top view of the structure at a fabrication stage of the processing method subsequent to FIGS. 3, 4.

(7) FIG. 6 is a cross-sectional view taken generally along line 6-6 in FIG. 5.

(8) FIG. 7 is a cross-sectional view of a structure in accordance with alternative embodiments of the invention.

(9) FIG. 8 is a cross-sectional view of a structure in accordance with alternative embodiments of the invention.

(10) FIG. 9 is a top view of a structure at an initial fabrication stage of a processing method in accordance with embodiments of the invention.

(11) FIG. 10 is a cross-sectional view taken generally along line 10-10 in FIG. 9.

(12) FIG. 10A is a cross-sectional view taken generally along line 10A-10A in FIG. 9.

(13) FIG. 11 is a top view of the structure at a fabrication stage of the processing method subsequent to FIGS. 9, 10, 10A.

(14) FIG. 12 is a cross-sectional view taken generally along line 12-12 in FIG. 11.

(15) FIG. 12A is a cross-sectional view taken generally along line 12A-12A in FIG. 11.

(16) FIG. 13 is a cross-sectional view of a structure in accordance with alternative embodiments of the invention.

(17) FIG. 14 is a cross-sectional view of a structure in accordance with alternative embodiments of the invention.

(18) FIG. 15 is a cross-sectional view of a structure in accordance with alternative embodiments of the invention.

(19) FIG. 16 is a top view of a structure in accordance with alternative embodiments of the invention.

(20) FIG. 17 is a top view of the structure at a fabrication stage of the processing method subsequent to FIG. 16.

(21) FIG. 18 is a cross-sectional view taken generally along line 18-18 in FIG. 17.

DETAILED DESCRIPTION

(22) With reference to FIGS. 1, 2 and in accordance with embodiments of the invention, a structure 10 for a photonics chip includes a waveguide core 12 that is disposed on, and over, a dielectric layer 14 and a substrate 16. In an embodiment, the dielectric layer 14 may be comprised of a dielectric material, such as silicon dioxide, and the substrate 16 may be comprised of a semiconductor material, such as single-crystal silicon. In an embodiment, the dielectric layer 14 may be a buried oxide layer of a silicon-on-insulator substrate. The waveguide core 12 is separated from the substrate 16 by the dielectric material of the intervening dielectric layer 14. In an alternative embodiment, one or more additional dielectric layers comprised of, for example, silicon dioxide may be disposed between the waveguide core 12 and the upper surface of the dielectric layer 14.

(23) In an embodiment, the waveguide core 12 may be comprised of a material having a refractive index that is greater than the refractive index of silicon dioxide. In an embodiment, the waveguide core 12 may be comprised of a semiconductor material, such as single-crystal silicon, amorphous silicon, or polysilicon. In an alternative embodiment, the waveguide core 12 may be comprised of a dielectric material, such as silicon nitride, silicon oxynitride, or aluminum nitride. In alternative embodiments, other materials, such as a III-V compound semiconductor, may be used to form the waveguide core 12.

(24) In an embodiment, the waveguide core 12 may be formed by patterning a layer with lithography and etching processes. In an embodiment, an etch mask may be formed by a lithography process over the layer, and unmasked sections of the layer may be etched and removed with an etching process. In an embodiment, the waveguide core 12 may be formed by patterning the semiconductor material (e.g., single-crystal silicon) of the device layer of a silicon-on-insulator substrate. In an embodiment, the waveguide core 12 may be formed by patterning a deposited layer comprised of its constituent material (e.g., silicon nitride). The waveguide core 12 may extend lengthwise along a longitudinal axis 15. The waveguide core 12 may have a width W1 in a direction transverse to the longitudinal axis 15.

(25) With reference to FIGS. 3, 4 in which like reference numerals refer to like features in FIGS. 1, 2 and at a subsequent fabrication stage, a dielectric layer 18 may be formed over the waveguide core 12. The dielectric layer 18 may be comprised of a dielectric material, such as silicon dioxide. The waveguide core 12 may be embedded in the dielectric material of the dielectric layer 18.

(26) A metal layer 20 may be formed on, and over, the dielectric layer 18. In an embodiment, the metal layer 20 may be comprised of a metal, such as copper or aluminum, that is compatible with complementary-metal-oxide-semiconductor processes. In an alternative embodiment, the metal layer 20 may be comprised of a metal, such as gold or silver, that is incompatible with

complementary-metal-oxide-semiconductor processes. In an embodiment, the metal layer **20** may be deposited and patterned by lithography and etching processes. In an embodiment, the metal layer **20** may have a width W_2 that is greater than the width W_1 of the waveguide core **12**.

(27) With reference to FIGS. **5**, **6** in which like reference numerals refer to like features in FIGS. **3**, **4** and at a subsequent fabrication stage, a dielectric layer **22** may be formed on, and over, the metal layer **20**. The dielectric layer **22** may be comprised of a dielectric material, such as silicon dioxide. The metal layer **20** may be embedded in the dielectric material of the dielectric layer **22**.

(28) A layer **24** may be disposed on the dielectric layer **22**. The layer **24** may be carried on a dielectric layer **26** and a substrate **28**. In an embodiment, the dielectric layer **26** may be comprised of a dielectric material, such as silicon dioxide, having a refractive index that is less than the refractive index of the layer **24**, and the substrate **28** may be comprised of silicon, lithium niobate, quartz, or fused silica. The layer **24** may be bonded to the dielectric layer **22** by a bonding process, such as dielectric bonding or hybrid bonding. The layer **24** may adjoin the dielectric layer **22** along a bonding interface **25** that is disposed between the layer **24** and the dielectric layer **22**. The layer **24** may be formed on the dielectric layer **26** and substrate **28**, and the substrate **28** may be manipulated to place the layer **24** in a contacting relationship with the dielectric layer **22** in advance of the bonding process. In an embodiment, after the contacting relationship is established, a low-temperature thermal anneal may be performed at a sufficient temperature and for a sufficient duration to create a face-to-face bond along the bonding interface **25** between the contacting surfaces of the dielectric layer **22** and the layer **24**. In an embodiment, the bonding interface **25** may lack any additional material, such as an adhesive, to promote the bonding between the surface of the dielectric layer **22** and the surface of the layer **24**.

(29) In an embodiment, the layer **24** may be comprised of a material that exhibits an electric-field-induced Pockels effect in which the refractive index varies in proportional to the strength of an applied electric field according to an electro-optic coefficient. In an embodiment, the layer **24** may be comprised of a crystalline material that lacks inversion symmetry and that is characterized by an optic axis whose refractive index is controllable by an applied electric field. In an embodiment, the layer **24** may be comprised of lithium niobate. In an embodiment, the layer **24** may be comprised of thin film lithium niobate having a thickness in a range of 300 nanometers to 900 nanometers. In alternative embodiments, the layer **24** may be comprised of lithium tantalate, lithium niobate doped with magnesium oxide, or barium titanate. In alternative embodiments, the layer **24** may be comprised of a binary or ternary III-V compound semiconductor material, such as gallium nitride, indium gallium nitride, indium phosphide, indium gallium arsenide, gallium arsenide, indium arsenide, or indium gallium phosphide. The layer **24** may have a width dimension W_3 that is greater than the width dimension W_1 of the waveguide core **12** and that is greater than the width dimension W_2 of the metal layer **20**.

(30) The dielectric material of the dielectric layer **22** separates the layer **24** from the metal layer **20** and prevents direct contact therebetween. The combination of the layer **24**, the metal layer **20**, and the intervening dielectric material of the dielectric layer **22** defines a hybrid plasmonic waveguide. In that regard, the proximity of the layer **24** and the metal layer **20** provides a mode supported by the layer **24** and a separate mode supported at the adjacent surface of the metal layer **20**. The dielectric material of the dielectric layer **18** separates the waveguide core **12** from the metal layer **20** and prevents direct contact therebetween. The combination of the waveguide core **12**, the metal layer **20**, and the intervening dielectric material of the dielectric layer **22** defines a hybrid plasmonic waveguide. In that regard, the proximity of the waveguide core **12** and the metal layer **20** provides a mode supported by the waveguide core **12** and a separate mode supported at the adjacent surface of the metal layer **20**.

(31) With reference to FIG. **7** and in accordance with alternative embodiments, the width dimension W_2 of the metal layer **20** may be increased to be, for example, equal to the width dimension W_3 of the layer **24** such that the metal layer **20** and the layer **24** are coterminous. The

dielectric layer 22 may be disposed in its entirety on, and over, metal layer 20.

(32) With reference to FIG. 8 and in accordance with alternative embodiments, a dielectric layer 27 may be formed on the layer 24 before the bonding process is performed. As a result, the bonding interface 25 is formed between the contacting surfaces of the dielectric layer 22 and the dielectric layer 27, and the layer 24 is separated from the metal layer 20 by the dielectric material of the dielectric layer 27 in addition to the dielectric material of the dielectric layer 22.

(33) With reference to FIGS. 9, 10, 10A and in accordance with alternative embodiments, a structure 30 for a photonics chip includes an edge coupler 32, an input optical coupler 34 connected to the edge coupler 32, an output optical coupler 36, and waveguide cores 38, 40 defining arms that are separately routed from the input optical coupler 34 to the output optical coupler 36, and an edge coupler 42 connected to the output optical coupler 36. The input optical coupler 34, output optical coupler 36, and waveguide cores 38, 40 define portions of a modulator 37. The edge coupler 32, input optical coupler 34, output optical coupler 36, waveguide cores 38, 40, and edge coupler 42 may be disposed on, and over, the dielectric layer 14 and the substrate 16. In an alternative embodiment, may one or more additional dielectric layers comprised of, for example, silicon dioxide may be positioned between the edge coupler 32, input optical coupler 34, output optical coupler 36, waveguide cores 38, 40, and edge coupler 42 and the top surface of the dielectric layer 14.

(34) In an embodiment, the edge coupler 32, input optical coupler 34, output optical coupler 36, waveguide cores 38, 40, and edge coupler 42 may be comprised of a material having a refractive index that is greater than the refractive index of silicon dioxide. In an embodiment, the edge coupler 32, input optical coupler 34, output optical coupler 36, waveguide cores 38, 40, and edge coupler 42 may be comprised of a semiconductor material, such as single-crystal silicon, amorphous silicon, or polysilicon. In an alternative embodiment, the edge coupler 32, input optical coupler 34, output optical coupler 36, waveguide cores 38, 40, and edge coupler 42 may be comprised of a dielectric material, such as silicon nitride, silicon oxynitride, or aluminum nitride. In alternative embodiments, other materials, such as a III-V compound semiconductor, may be used to form the edge coupler 32, input optical coupler 34, output optical coupler 36, waveguide cores 38, 40, and edge coupler 42.

(35) In an embodiment, the edge coupler 32, input optical coupler 34, output optical coupler 36, waveguide cores 38, 40, and edge coupler 42 may be formed by patterning a layer with lithography and etching processes. In an embodiment, an etch mask may be formed by a lithography process over the layer, and unmasked sections of the layer may be etched and removed with an etching process. The masked sections of the layer may determine the patterned shapes of the edge coupler 32, input optical coupler 34, output optical coupler 36, waveguide cores 38, 40, and edge coupler 42. In an embodiment, the edge coupler 32, input optical coupler 34, output optical coupler 36, waveguide cores 38, 40, and edge coupler 42 may be formed by patterning the semiconductor material (e.g., single-crystal silicon) of the device layer of a silicon-on-insulator substrate. In an embodiment, the edge coupler 32, input optical coupler 34, output optical coupler 36, waveguide cores 38, 40, and edge coupler 42 may be formed by patterning a deposited layer comprised of their constituent material (e.g., silicon nitride). In an alternative embodiment, each of the waveguide cores 38, 40 may be slotted.

(36) The input optical coupler 34 has an input port coupled to the edge coupler 32 and output ports respectively coupled to the waveguide cores 38, 40. The input optical coupler 34 splits the input light supplied from the edge coupler 32 between the waveguide core 38 and the waveguide core 40. In an embodiment, the input optical coupler 34 may be a 50-50 directional coupler that splits the input light evenly between the waveguide cores 38, 40. The output optical coupler 36 has an output port coupled to the edge coupler 42 and input ports coupled to the waveguide cores 38, 40. The output optical coupler 36 combines the light received from the waveguide cores 38, 40 and may also be a 50-50 directional coupler to match the input optical coupler 34. In an embodiment, the

input optical coupler **34** and output optical coupler **36** may be configured as multi-mode interference couplers each having a multi-mode interference region.

(37) The edge coupler **32** may include a tapered section **33** of a waveguide core that is connected to the input optical coupler **34**. The tapered section **33** of the edge coupler **32** terminates at a facet **35**. The tapered section **33** may have a width dimension that increases with increasing distance from the facet **35**. In an embodiment, the width dimension of the tapered section **33** may increase linearly with increasing distance from the facet **35**. In an alternative embodiment, the width dimension of the tapered section **33** may vary based on a non-linear function, such as a quadratic function, a cubic function, a parabolic function, a sine function, a cosine function, a Bezier function, or an exponential function. In an embodiment, the tapered section **33** may include a single stage of tapering characterized by a taper angle. In an alternative embodiment, the tapered section **33** may taper in multiple stages each characterized by a different taper angle.

(38) The edge coupler **42** may include a tapered section **43** of a waveguide core that is connected to the input optical coupler **34**. The tapered section **43** of the edge coupler **42** terminates at a facet **45**. The tapered section **43** may have a width dimension that increases with increasing distance from the facet **45**. In an embodiment, the width dimension of the tapered section **43** may increase linearly with increasing distance from the facet **45**. In an alternative embodiment, the width dimension W_2 of the tapered section **43** may vary based on a non-linear function, such as a quadratic function, a cubic function, a parabolic function, a sine function, a cosine function, a Bezier function, or an exponential function. In an embodiment, the tapered section **43** may include a single stage of tapering characterized by a taper angle. In an alternative embodiment, the tapered section **43** may taper in multiple stages each characterized by a different taper angle.

(39) In alternative embodiment, the edge couplers **32**, **42** may have a different configuration. For example, the edge couplers **32**, **42** may be divided into segments. As another example, the edge couplers **32**, **42** may include sections that are arranged to provide multiple tips. As another example, the edge couplers **32**, **42** may include sections that are arranged in different levels.

(40) The waveguide cores **38**, **40** include bends adjacent to the input optical coupler **34** that increase the separation between the waveguide cores **38**, **40** and decrease crosstalk between the arms of the modulator **37**. The waveguide cores **38**, **40** include bends adjacent to the output optical coupler **36** that decrease the separation between the waveguide cores **38**, **40** to converge at the output optical coupler **36**.

(41) With reference to FIGS. **11**, **12**, **12A** in which like reference numerals refer to like features in FIGS. **9**, **10**, **10A** and at a subsequent fabrication stage, the dielectric layer **18**, the metal layer **20**, and the dielectric layer **22** may be formed over the edge coupler **32**, input optical coupler **34**, output optical coupler **36**, waveguide cores **38**, **40**, and edge coupler **42**. The layer **24** may be bonded to the dielectric layer **22** by a bonding process that establishes the bonding interface **25** between the layer **24** and the dielectric layer **22** such that the layer **24**, dielectric layer **26**, and substrate **28** are bonded to the dielectric layer **22**.

(42) The structure **30** may include an electrode **48**, an electrode **50**, and an electrode **52** that is disposed with a juxtaposed relationship between the electrode **48** and the electrode **50**. The electrodes **48**, **50**, **52** may be positioned on a portion of the layer **24** inside a cavity **49** that is patterned by lithography and etching processes in the dielectric layer **26** and substrate **28**. The electrodes **48**, **50**, **52** may be comprised of a metal, such as gold, that is deposited and patterned by lithography and etching processes. In an embodiment, the metal layer **20** may be comprised of a different metal than the electrodes **48**, **50**, **52**. In an alternative embodiment, the metal layer **20** may be comprised of the same metal as the electrodes **48**, **50**, **52**. The modulator **37** further includes the electrodes **48**, **50**, **52**, the portion of the layer **24** on which the electrodes **48**, **50**, **52** are located, and the metal layer **20**, in addition to the input optical coupler **34**, the output optical coupler **36**, and waveguide cores **38**, **40**.

(43) In an embodiment, the cavity **49** may extend to the layer **24** and the electrodes **48**, **50**, **52** may

directly contact the portion of the layer **24** inside the cavity **49**. In an embodiment, the cavity **49** may extend partially through the dielectric layer **26** and the electrodes **48**, **50**, **52** may contact the residual thickness of the dielectric layer **26**. An underlying portion of the waveguide core **38** is disposed in a lateral direction between the electrode **48** and the electrode **52**. An underlying portion of the waveguide core **40** is disposed in a lateral direction between the electrode **50** and the electrode **52**. The electrode **52** may be coupled to a signal source, such as a driver that is configured to supply radiofrequency signals to the electrode **52** in order to modulate the light propagating in the waveguide cores **38**, **40** of the modulator **37**. The electrode **48** and the electrode **50** may be coupled to electrical ground.

(44) A light source **54** may be placed adjacent to the edge coupler **32**. In an embodiment, the light source **54** includes a light output **56** that is aimed toward the facet **35** of the tapered section **33** of the edge coupler **32** and that is configured to provide light in a mode propagation direction toward the edge coupler **32**. In an embodiment, the light source **54** may be a laser chip that includes a semiconductor laser configured to emit light (e.g., laser light) of a given wavelength, intensity, mode shape, and mode size from the light output **56** that is coupled to the edge coupler **32**. In an embodiment, the light source **54** may output coherent light in an infrared wavelength range. In an alternative embodiment, the light source **54** may be a single-mode or multi-mode optical fiber.

(45) The light source **54** may be disposed inside a cavity **55** in the substrate **16** that is formed adjacent to the facet **35** of the edge coupler **32**. In an alternative embodiment, an undercut may extend in the substrate **16** from the cavity **55** beneath the edge coupler **32** such that a portion of the dielectric layer **14** defines a membrane that supports the edge coupler **32** over the undercut. In an alternative embodiment, a sealed undercut may be provided in the substrate **16** beneath the waveguide cores **38**, **40** of the modulator **37**.

(46) An optical fiber **58** may be placed adjacent to the edge coupler **42**. The optical fiber **58** is configured to receive modulated light output from the facet **45** of the tapered section **43** of the edge coupler **42** and to route the modulated light away from the edge coupler **42**. For example, the optical fiber **58** may route the modulated light output from the edge coupler **42** over a long distance to a remote receiver. The optical fiber **58** may be disposed inside a groove **57** in the substrate **16** that is formed adjacent to the facet **45** of the edge coupler **42**. In an alternative embodiment, an undercut may extend in the substrate **16** from the groove **57** beneath the edge coupler **42** such that a portion of the dielectric layer **14** defines a membrane that supports the edge coupler **42** over the undercut.

(47) In an embodiment, the metal layer **20** and the layer **24** may fully overlap with the edge coupler **32**, input optical coupler **34**, output optical coupler **36**, waveguide cores **38**, **40**, and edge coupler **42**. In an embodiment, the metal layer **20** and the layer **24** may have a side edge **60** that is disposed adjacent to the facet **35** at the end of the tapered section **33** of the edge coupler **32**. In an embodiment, the metal layer **20** and the layer **24** may have a side edge **62**, which is opposite from the side edge **60**, that is disposed adjacent to the facet **45** at the end of the tapered section **43** of the edge coupler **42**. In an embodiment, the side edge **60** and the facet **35** may be coterminous. In an embodiment, the side edge **60** and the facet **35** may be substantially coterminous. In an embodiment, the side edge **62** and the facet **45** may be coterminous. In an embodiment, the side edge **62** and the facet **45** may be substantially coterminous.

(48) The structure **30** may define a transmitter in which the modulator **37** is configured to modulate light, such as continuous infrared light, received from the light source **54** via the edge coupler **32** and to output the modulated light via the edge coupler **42** to the optical fiber **58** for transmission to a remote location. In use, the edge coupler **32** may receive light from the light source **54** as input to the structure **30**. The input optical coupler **34** splits the input light between the waveguide cores **38**, **40**. A radiofrequency signal may be applied as a highspeed data signal to the electrode **52**. The output optical coupler **36** combines the light exiting the waveguide cores **38**, **40** to provide either constructive or destructive interference according to the value of the phase difference. Light may

be output from the output optical coupler **36** to the edge coupler **42** due to constructive interference if the phase delay is zero or not output due to destructive interference if the phase difference is equal to one-half wavelength.

(49) The inclusion of the metal layer **20** in the modulator **37** may enable the formation of the modulator **37**, which includes the layer **24**, on a monolithic platform. The modulator **37** may exhibit an improved modulation efficiency with strengthened light confinement, a minimized insertion loss, and a lowered propagation loss due to the inclusion of the metal layer **20**. The dielectric material of the dielectric layer **22** separates the layer **24** from the metal layer **20** and prevents direct contact therebetween. The combination of the metal layer **20**, the layer **24**, and the intervening dielectric material of the dielectric layer **22** defines a hybrid plasmonic waveguide. The dielectric material of the dielectric layer **18** separates each of the waveguide cores **38**, **40** from the metal layer **20** and prevents direct contact therebetween. The combinations of the waveguide cores **38**, **40**, the metal layer **20**, and the intervening dielectric material of the dielectric layer **18** define respective hybrid plasmonic waveguides.

(50) With reference to FIG. **13** and in accordance with alternative embodiments, the layer **24** may include ridges **46** that are disposed inside the cavity **49** between the electrodes **48**, **50**, **52**. One of the ridges **46** may be disposed between the electrode **48** and the electrode **50**, and the other of the ridges **46** may be disposed between the electrode **52** and the electrode **50**. In an embodiment, one of the ridges **46** may overlap with a portion of the waveguide core **38**, and one of the ridges **46** may overlap with a portion of the waveguide core **40**. In an embodiment, the ridges **46** may extend from a side edge of the cavity **49** adjacent to the input optical coupler **34** to a side edge of the cavity **49** adjacent to the output optical coupler **36**.

(51) With reference to FIG. **14** and in accordance with alternative embodiments, the metal layer **20** may include slots **47** that introduce discontinuities. The slots **47** may be formed by patterning the metal layer **20** with lithography and etching processes. In an embodiment, each slot **47** may overlap with a portion of one of the waveguide cores **38**, **40**. In an embodiment, each ridge **46** in the layer **24** may overlap at least in part with one of the slots **47**.

(52) With reference to FIG. **15** and in accordance with alternative embodiments, the layer **24** may include slots **51** that introduce discontinuities. The slots **51** may be formed by patterning the layer **24** with lithography and etching processes. In an embodiment, each slot **51** may overlap with a portion of one of the waveguide cores **38**, **40**.

(53) With reference to FIG. **16** and in accordance with alternative embodiments, a structure **70** may include a waveguide core **72** that is disposed between the edge coupler **32** and the edge coupler **42**, as well as a ring resonator **74** that is positioned adjacent to a portion of the waveguide core **72**. In an embodiment, the ring resonator **74** may be separated from the adjacent portion of the waveguide core **72** by a gap. In an embodiment, the ring resonator **74** may be a waveguide core that is ring shaped or round. In an alternative embodiment, the ring resonator **74** may be a waveguide core that is racetrack shaped or oblong.

(54) With reference to FIGS. **17**, **18** in which like reference numerals refer to like features in FIG. **16** and at a subsequent fabrication stage, electrodes **76**, **78** are formed on the layer **24**. The electrodes **76**, **78** may have truncated ring shapes with the underlying ring resonator **74** disposed between the electrode **76** and the electrode **78**. The electrodes **76**, **78** are disposed in a different elevation level of the structure **70** than the ring resonator **74** with the ring resonator **74** disposed in a vertical direction between the electrodes **76**, **78** and the substrate **16**. In an embodiment, the electrode **76** may be disposed outside the outer perimeter of the underlying ring resonator **74**, and the electrode **78** may be disposed inside the inner perimeter of the underlying ring resonator **74**. In an embodiment, the electrode **76** may be disposed outside the outer circumference of the underlying ring resonator **74**, and the electrode **78** may be disposed inside the inner circumference of the underlying ring resonator **74**. The ring resonator **74**, the electrodes **76**, **78**, and the portion of the layer **24** between the ring resonator **74** and electrodes **76**, **78** define a modulator **77** that can be

used to switch the light propagating in the waveguide core 72. In an embodiment, the metal layer 20 may have an overlapping relationship with the edge couplers 32, 42, the waveguide core 72, and the ring resonator 74.

(55) In use, light (e.g., laser light) propagating in the waveguide core 72 may be coupled and transferred in a lateral direction from the waveguide core 72 to the ring resonator 74. The electrode 76 may be coupled to a signal source, such as a driver that includes a driver amplifier that is configured to supply radiofrequency signals to the electrode 76 in order to modulate the light propagating in the waveguide core of the ring resonator 74. The electrode 78 may be coupled to electrical ground.

(56) The modulator 77 may exhibit an improved modulation efficiency with strengthened light confinement, a minimized insertion loss, and a lowered propagation loss due to the inclusion of the metal layer 20.

(57) The methods as described above are used in the fabrication of integrated circuit chips. The resulting integrated circuit chips can be distributed by the fabricator in raw wafer form (e.g., as a single wafer that has multiple unpackaged chips), as a bare die, or in a packaged form. The chip may be integrated with other chips, discrete circuit elements, and/or other signal processing devices as part of either an intermediate product or an end product. The end product can be any product that includes integrated circuit chips, such as computer products having a central processor or smartphones.

(58) References herein to terms modified by language of approximation, such as “about”, “approximately”, and “substantially”, are not to be limited to the precise value specified. The language of approximation may correspond to the precision of an instrument used to measure the value and, unless otherwise dependent on the precision of the instrument, may indicate a range of $\pm 10\%$ of the stated value(s).

(59) References herein to terms such as “vertical”, “horizontal”, etc. are made by way of example, and not by way of limitation, to establish a frame of reference. The term “horizontal” as used herein is defined as a plane parallel to a conventional plane of a semiconductor substrate, regardless of its actual three-dimensional spatial orientation. The terms “vertical” and “normal” refer to a direction in the frame of reference perpendicular to the horizontal, as just defined. The term “lateral” refers to a direction in the frame of reference within the horizontal plane.

(60) A feature “connected” or “coupled” to or with another feature may be directly connected or coupled to or with the other feature or, instead, one or more intervening features may be present. A feature may be “directly connected” or “directly coupled” to or with another feature if intervening features are absent. A feature may be “indirectly connected” or “indirectly coupled” to or with another feature if at least one intervening feature is present. A feature “on” or “contacting” another feature may be directly on or in direct contact with the other feature or, instead, one or more intervening features may be present. A feature may be “directly on” or in “direct contact” with another feature if intervening features are absent. A feature may be “indirectly on” or in “indirect contact” with another feature if at least one intervening feature is present. Different features “overlap” if a feature extends over, and covers a part of, another feature.

(61) The descriptions of the various embodiments of the present invention have been presented for purposes of illustration but are not intended to be exhaustive or limited to the embodiments disclosed. Many modifications and variations will be apparent to those of ordinary skill in the art without departing from the scope and spirit of the described embodiments. The terminology used herein was chosen to best explain the principles of the embodiments, the practical application or technical improvement over technologies found in the marketplace, or to enable others of ordinary skill in the art to understand the embodiments disclosed herein.

Claims

1. A structure for a photonics chip, the structure comprising: a first substrate; a second substrate; a first waveguide core on the first substrate; a first layer that has an overlapping relationship with the first waveguide core, the first layer comprising a first metal; a second layer that has an overlapping relationship with the first waveguide core and the first layer, the second layer comprising a material that exhibits an electro-optic effect; a first dielectric layer on the second substrate, the second layer disposed on the first dielectric layer; and a second dielectric layer between the first layer and the second layer, the second dielectric layer adjoining the second layer along a bonding interface.
2. The structure of claim 1 further comprising: a second waveguide core on the first substrate, wherein the first layer has an overlapping relationship with the second waveguide core.
3. The structure of claim 2 further comprising: an input optical coupler; and an output optical coupler, wherein the first waveguide core and the second waveguide core are coupled to the input optical coupler, and the first waveguide core and the second waveguide core are coupled to the output optical coupler.
4. The structure of claim 3 further comprising: a first edge coupler connected by the input optical coupler to the first waveguide core; and a second edge coupler connected by the output optical coupler to the second waveguide core, wherein the first layer has an overlapping relationship with the first edge coupler and the second edge coupler.
5. The structure of claim 4 further comprising: a light source configured to supply light to the first edge coupler; and an optical fiber configured to receive the light from the second edge coupler.
6. The structure of claim 4 wherein the second layer has an overlapping relationship with the first edge coupler and the second edge coupler.
7. The structure of claim 2 wherein the second layer includes a first ridge that overlaps with the first waveguide core, and the second layer includes a second ridge that overlaps with the second waveguide core.
8. The structure of claim 2 further comprising: an electrode on the second layer, the electrode disposed in a lateral direction between a portion of the first waveguide core and a portion of the second waveguide core, wherein the first layer and the second layer are disposed between the electrode and the first waveguide core.
9. The structure of claim 8 wherein the electrode comprises a second metal different from the first metal of the first layer.
10. The structure of claim 2 wherein the second layer includes a first slot that overlaps with the first waveguide core, and the second layer includes a second slot that overlaps with the second waveguide core.
11. The structure of claim 2 wherein the first layer includes a first slot that overlaps with the first waveguide core, and the first layer includes a second slot that overlaps with the second waveguide core.
12. The structure of claim 1 further comprising: a ring resonator on the first substrate adjacent to a portion of the first waveguide core; a first electrode on the second layer; and a second electrode on the second layer, wherein the ring resonator includes a portion disposed in a lateral direction between the first electrode and the second electrode, and the first layer and the second layer overlap with the ring resonator.
13. The structure of claim 1 wherein the material is lithium niobate, lithium tantalate, lithium niobate doped with magnesium oxide, or barium titanate.
14. The structure of claim 1 wherein the material is a III-V compound semiconductor.
15. A structure for a photonics chip, the structure comprising: a substrate; a waveguide core on the substrate; a first layer that has an overlapping relationship with the waveguide core, the first layer comprising a metal; a second layer that has an overlapping relationship with the waveguide core and the first layer, the second layer comprising a material that exhibits an electro-optic effect; a first dielectric layer between the first layer and the second layer; and a second dielectric layer

between the first dielectric layer and the second layer, wherein the second dielectric layer adjoins the first dielectric layer along a bonding interface.

16. A method of forming a structure for a photonics chip, the method comprising: forming a waveguide core on a first substrate; forming a first layer that has an overlapping relationship with the waveguide core, wherein the first layer comprises a metal; forming a second dielectric layer over the first layer; and forming a second layer that has an overlapping relationship with the waveguide core and the first layer, wherein the second layer comprises a material that exhibits an electro-optic effect, wherein the second layer is formed on a second substrate, and the second layer is bonded to the second dielectric layer along a bonding interface between the first substrate and the second substrate.
